



# Heavy Metals and Magnetic Susceptibility Signature in *Muntingia calabura* as Proxy Indicator of Mining Area Pollution Level

Siti Zulaikah<sup>1\*</sup>, Dewi Nur Alfiah<sup>1</sup>, Asfianti Latifah<sup>1</sup>, Mochammad Bagas Setya Rahman<sup>1</sup>, Cahyo Aji Hapsoro<sup>1</sup>, Ardyanto Tanjung<sup>2</sup>, Ann Marie Hirt<sup>3</sup>, Hanif 'Izzuddin Zakly<sup>1</sup>, Muhammad Fathur Rouf Hasan<sup>4</sup>

<sup>1</sup> Department of Physics, Faculty of Mathematic and Natural Sciences, Universitas Negeri Malang, 65145 Malang, Indonesia

<sup>2</sup> Department of Geography, Faculty of Social Sciences, Universitas Negeri Malang, 65145 Malang, Indonesia

<sup>3</sup> Institute of Geophysics, ETH-Zurich, CH-8092 Zurich, Switzerland

<sup>4</sup> Graduate School, Universitas Brawijaya, 65145 Malang, Indonesia

\* Correspondence: Siti Zulaikah ([siti.zulaikah.fmipa@um.ac.id](mailto:siti.zulaikah.fmipa@um.ac.id))

Received: 05-27-2025

Revised: 10-25-2025

Accepted: 11-21-2025

**Citation:** S. Zulaikah, D. N. Alfiah, A. Latifah, M. B. S. Rahman, C. A. Hapsoro, A. Tanjung, A. M. Hirt, H. 'I. Zakly, and M. F. R. Hasan, "Heavy metals and magnetic susceptibility signature in *Muntingia calabura* as proxy indicator of mining area pollution level," *Int. J. Environ. Impacts.*, vol. 9, no. 1, pp. 188–199, 2026. <https://doi.org/10.56578/ije090115>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

**Abstract:** Heavy metals tracing and magnetic susceptibility are generally used as a proxy indicator of pollution in various depositional environments. This research focused on tracing the significant record of pollution of the *Muntingia calabura* tree to understand the sensitivity in recording pollution and mining production. The area of this research is a hydrocarbon mining site of Wonocolo Geopark in the Bojonegoro, East Java Indonesia. The samples were taken both polluted and non-polluted leaf and bark, from 20 sampling points. Polluted leaves then were characterized by the existence of elevated levels of Pb, Fe, Cu and Zn. The average magnetic susceptibility of leaves increases from  $0.86 \times 10^{-8} \text{ m}^3/\text{kg}$  in non-polluted samples to  $13.55 \times 10^{-8} \text{ m}^3/\text{kg}$  in polluted samples and in the same way, increasing of magnetic susceptibility was also seen in the barks, from an average of  $0.21 \times 10^{-8} \text{ m}^3/\text{kg}$  in non-polluted sample, to  $2.55 \times 10^{-8} \text{ m}^3/\text{kg}$  in polluted sample. The pattern of magnetic susceptibility on leaves and barks at each sampling point is also the same as the pattern of hydrocarbon production which is related to the level of pollution in the area. The increase of magnetic susceptibility in polluted leaves and barks is thought caused by input of magnetic minerals and heavy metals from the fly ash of diesel engines used for hydrocarbon mining process. The heavy metal concentration has the average of Pb (0.070 ppm), Fe (13.322 ppm), Cu (8.434 ppm), and Zn (11.668 ppm). This value has exceeded the threshold of heavy metals content and have a worst impact on health and the environment. Based on Pollution Load Index (PLI) calculations, the most of areas affected by heavy metal pollution in very high and extremely high levels with the highest pollutants input are Fe, Cu, Zn and Pb respectively.

**Keywords:** Magnetic susceptibility; Proxy indicator; Heavy metals; Mining product; Pollution record; *Muntingia calabura*

## 1 Introduction

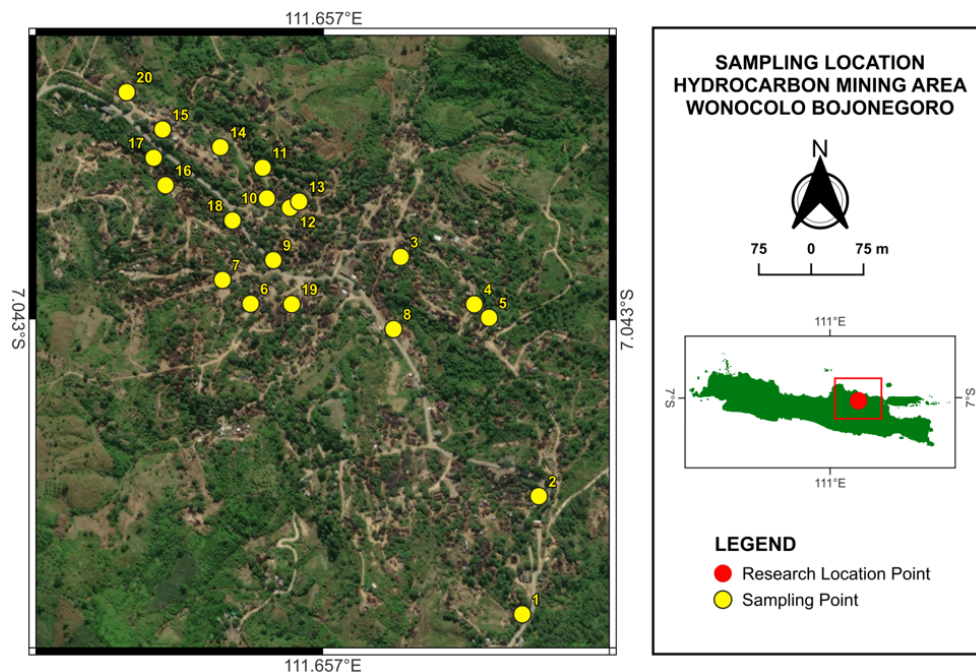
A recent report by the World Health Organization (WHO) outlines the deadly effects of ambient (air) pollution, which leads to increase rates of acute and chronic respiratory disease, strokes, heart disease, and lung cancer. Air pollution is ascribed to have caused an 4.2 million premature deaths in 2019 and approximately 89% of these occurred in Southeast Asia and the Western Pacific [1]. In Indonesia the number of premature deaths is estimated to be 119,504 in 2019 [2]. It is well documented that fine particulate matter (PM) finds their way through inhalation into every organ of the human body. Metallic within the body is not only associated with disease, but also can lead to deterioration of cognitive function and neurological development [3–9].

Numerous studies have shown that Fe-bearing nanoparticles are very common in the air [10, 11]. Airborne Fe-bearing nanoparticles are particularly hazardous to health, because Fe is found in metalloproteins in the body that play a role in neuronal functions [4]. For this reason, rock magnetic methods have been successfully applied in assessing air pollution [3, 12–15]. Plants have been widely used for biomonitoring of air pollution [16–24], because

their surfaces act as a trap for PM [25]. Different plants or parts of plants can be used as samples for biomonitoring, for example leaves [4, 5, 8, 9], bark [5, 9, 26, 27], tree rings [28–30], and lichen [31]. These studies have demonstrated that the magnetic properties of plants, e.g., magnetic susceptibility and saturation isothermal remanent magnetization (SIRM), are enhanced in industrial areas compared to residential or ecological areas.

One reason for using magnetic properties for biomonitoring pollution, is that the measurements take relatively little time, which allows for a large number of samples to be evaluated efficiently [32, 33]. By combining magnetic data with geochemical information, it is possible to link a magnetic parameter with heavy metal concentration [32, 34–39]. Although no universal correlations can be established between specific magnetic parameters and specific heavy metals, valid correlations can be established within a study area. Thus, once a relation is established, magnetic properties can be used to obtain detail maps for specific metal concentrations.

Magnetic susceptibility has been used successfully to monitor the level of hydrocarbon pollution in soils [33, 40, 41]. These studies have also demonstrated that soil cores that were sampled at a distance from a well or oil distillation site had low susceptibility throughout the soil column. The Wonocolo hydrocarbon mining that also become the Geopark, located in the Bojonegoro Regency in East Java, Indonesia (Figure 1), is being developed for geology-based tourism. The park preserves traditional methods of oil extraction and production, which are still in use today. But this mining activity causes pollution in the mining area and has a big impact on the surrounding environment. So, the pollution level is very important to evaluate or monitor. A preliminary study of soil cores in the Wonocolo Geopark, Bojonegoro Indonesia showed high susceptibility in the upper soils in areas where there was active oil production either presently or in the past.



**Figure 1.** Sample location point in Wonocolo hydrocarbon mining, Bojonegoro-East Java

The purpose of this study is to evaluate the applicability of using rock magnetic methods to monitor the bioaccumulation of heavy metals associated with hydrocarbon wells in the Wonocolo Geopark. Both leaf and bark samples are sampled from *Muntingia calabura*, a small evergreen tree found in tropical climates. *Muntingia calabura* is a productive shade plant, with edible fruit. *Muntingia calabura* is widely used as a shade plant in several mining and residential areas. This plant is widely planted at the research site, and it is dominant at active well points. The use of *Muntingia calabura* as a plant sample is also based on Lin [42].

Elemental analysis of the biomaterial is also performed to determine the heavy metal concentration, and to correlate against magnetic properties. Our approach will allow us to compare the effectiveness of leaves versus bark in trapping heavy metal PM. Our results are used to map the area with respect to the spatial distribution of the heavy metals and magnetic susceptibility. The results from this study will be used to establish recommendations for using the *Muntingia calabura* plant as a method to contain PM locally [7, 8, 26].

## 2 Material and Method

Leaf and bark samples of *Muntingia calabura* were taken at 20 sites within the hydrocarbon site in the Wonocolo Geopark, Bojonegoro Indonesia. The sample sites were selected based on the location of active traditional hydrocarbon

mining wells in the area. Although not all sites were sampled, the samples taken represent a large and representative portion (Figure 1). *Muntingia calabura* has slightly fur leaves and a relatively smooth bark. Therefore, we will also examine its effectiveness as a bioaccumulator and which parts of the plant are effective as bioaccumulators, which will certainly differentiate it from other plants.

The sites were chosen based on the color of the leaves. Polluted samples were taken from locations directly facing the mining well activity, while unpolluted samples were taken from locations behind the mining activity. Unpolluted samples have green leaves and brown bark and with both becoming black in areas with notable polluted area (Figure 2). The polluted samples were taken from plants that are near to active wells, in addition to distillation fumes from the run pumps. These samples can be distinguished by the dark color of polluted sites and the light color of unpolluted sites.



**Figure 2.** (a) Difference in color for a polluted (black) and a non-polluted (green) leaf; (b) polluted bark is blackish and non-polluted bark is brown

These leaves are labeled DTPx for the leaf samples and BTPx for bark samples. Non-polluted samples were taken at places that are not exposed to ash pollution and where the leaf and bark colors are still fresh (Figure 2).

These are labeled DTTPx for the leaves and BTTPx for bark. After collection the samples were clipped together and placed into Plastic Bags stored in boxes and taken to the laboratory in Malang. The samples were first dried in the sun (outside), where the average daily temperature was 27 °C. The dried samples were then cut into small pieces and put into standard plastic magnetic measurement holders and weighed with a digital balance. The mass susceptibility of the samples was measured on a Bartington dual frequency MS2B susceptibility meter, using both frequencies: high frequencies (4700 Hz) and low frequency (470 Hz). The contaminated leaf samples (DTPx) were then subjected to evaluation with an atomic absorption spectroscopy (AAS) to determine the heavy metals content (Fe, Cu, Pb, and Zn). Only leaf samples were used to determine heavy metal content, as recommended by earlier research that leaves are better recorders of contamination compared to other parts of plants [43]. A further sample to serve as a reference for an unpolluted site that is away from any hydrocarbon activity or roads, it is from a residential area in Malang. Pollution Load Index (PLI) was calculated, using the formula,  $PLI = (C_{Fe} \times C_{Cu} \times C_{Pb} \times C_{Zn})^{(1/4)}$ . Statistical analysis was made using IBS SPSS software.

### 3 Result

The magnetic susceptibility of the leaf and bark is listed in Table 1. Magnetic susceptibility of leaves from less polluted areas ranges from  $-0.641 \times 10^{-8} \text{ m}^3/\text{kg}$  to  $2.481 \times 10^{-8} \text{ m}^3/\text{kg}$  with an average of  $0.856 \pm 0.944 \times 10^{-8} \text{ m}^3/\text{kg}$  for non-polluted samples and from  $1.387 \times 10^{-8} \text{ m}^3/\text{kg}$  to  $54.764 \times 10^{-8} \text{ m}^3/\text{kg}$  with an average of  $13.544 \pm 19.153 \times 10^{-8} \text{ m}^3/\text{kg}$  for polluted sample. Magnetic susceptibility of the bark samples is between  $-1.203 \times 10^{-8} \text{ m}^3/\text{kg}$  to  $2.260 \times 10^{-8} \text{ m}^3/\text{kg}$  with an average of  $0.211 \pm 0.903 \times 10^{-8} \text{ m}^3/\text{kg}$  for non-polluted samples and from  $0.147 \times 10^{-8} \text{ m}^3/\text{kg}$  to  $9.054 \times 10^{-8} \text{ m}^3/\text{kg}$  with an average of  $2.945 \pm 2.150 \times 10^{-8} \text{ m}^3/\text{kg}$  for polluted samples. The reference sample from a residential area was diamagnetic with a value of  $-0.745 \times 10^{-8} \text{ m}^3/\text{kg}$ . Because of bark susceptibility is low, we decided to not measure AAS. Furthermore, based on references, *Muntingia calabura*'s bark is classified as glabrous, making it ineffective at trapping airborne metals and the bark is always obscured by dense leaves, which sometimes hang down to the ground.

Results from elemental analysis, which includes Pb, Fe, Cu and Zn, is provided in Table 2. The elemental concentrations for the different samples are highly variable. Site 1 had very high Cu and Fe concentrations and low Zn concentration compared to the other sites. Fe concentrations range from 0.550 ppm to 71.019 ppm with an average  $13.322 \pm 15.101 \text{ ppm}$ , and Zn concentrations range from 1.087 ppm to 26.131 ppm with an average of  $11.668 \pm 6.883 \text{ ppm}$ . The concentration of Cu varied from 0.093 ppm to 26.131 ppm with an average of  $8.434 \pm 36.559 \text{ ppm}$ ; if

site 1 is excluded the average decreases by an order of magnitude to  $0.345 \pm 0.031$  ppm. The concentration of Pb is the lowest and ranges from non-detectable concentration to 0.249 ppm with an average of  $0.073 \pm 0.059$  ppm. The sample from the residential area has Fe, Cu, and Zn concentrations that are about an order of magnitude less than the average of the contaminated leaves. It is interesting to note, however, that the Pb concentration was on the same order as many of the polluted samples.

Heavy metal concentrations of Pb were found in leaves ranging from 0 ppm to 0.249 ppm with an average of 0.070 ppm. This value has exceeded the specified standard and according to WHO, lead levels in the air must be reduced to zero. Because lead is extremely dangerous when it enters the body in enormous quantities. Among the dangers of lead to the body are hypertension, miscarriages, brain injury, abdominal pain, peripheral nerve damage, sperm damage, cognitive impairment, etc [44].

**Table 1.** Magnetic susceptibility measurement ( $\chi_{lf}$ ) of non-polluted and polluted leaves and bark from *Muntingia calabura* in the bojonegoro hydrocarbon mining area

| Sample | Magnetic Susceptibility ( $\chi_{lf}$ ) ( $10^{-8} \text{ m}^3/\text{kg}$ ) |        |        |       |
|--------|-----------------------------------------------------------------------------|--------|--------|-------|
|        | DTTP                                                                        | DTP    | BTTP   | BTP   |
| 1      | -0.341                                                                      | 54.764 | 0.462  | 9.054 |
| 2      | -0.641                                                                      | 1.387  | -0.295 | 5.366 |
| 3      | 0.621                                                                       | 17.741 | -0.445 | 2.107 |
| 4      | 2.481                                                                       | 2.667  | 0.270  | 0.409 |
| 5      | -0.310                                                                      | 4.536  | -0.914 | 1.497 |
| 6      | 0.587                                                                       | 1.660  | -0.643 | 7.277 |
| 7      | 0.609                                                                       | 9.158  | -0.436 | 1.059 |
| 8      | 1.645                                                                       | 4.345  | 0.141  | 1.835 |
| 9      | 0.789                                                                       | 15.408 | 0.161  | 3.272 |
| 10     | 0.592                                                                       | 3.592  | 0.488  | 1.733 |
| 11     | 1.071                                                                       | 4.259  | 1.556  | 3.662 |
| 12     | 0.448                                                                       | 27.331 | -1.203 | 0.831 |
| 13     | -0.454                                                                      | 22.656 | 0.429  | 0.147 |
| 14     | 2.390                                                                       | 9.191  | 2.260  | 2.849 |
| 15     | 1.137                                                                       | 3.349  | 0.843  | 6.661 |
| 16     | 0.3183                                                                      | 32.440 | -0.874 | 3.591 |
| 17     | 0.754                                                                       | 11.227 | 0.664  | 3.457 |
| 18     | 1.029                                                                       | 1.6884 | 0.404  | 2.422 |
| 19     | 2.404                                                                       | 30.003 | 1.661  | 0.964 |
| 20     | 1.986                                                                       | 13.481 | -0.308 | 0.713 |
| Mean   | 0.856                                                                       | 13.544 | 0.211  | 2.945 |

Note: DTTP, non-polluted leaf samples; DTP, polluted leaf samples; BTTP, non-polluted bark samples; BTP, polluted bark samples.

Heavy metal concentrations of Fe were found in leaves ranging from 0.550 ppm to 71.019 ppm with an average of 13.322 ppm. This is related to anthropogenic contaminants that are a source of pollutants such as engine activity and metal absorption from tanks which allow them to burn as oxides. The Fe concentration far exceeds the standard Fe in the air, which is 0.438. If humans inhale much Fe, it can cause several complications in the body. In fact, research shows that population with higher levels of iron in the body has a higher risk of developing cancer [45]. Excess Fe can also cause neurological disorders [46].

The concentration of heavy metal Cu detected in the leaves was 0.093 ppm to 163.882 ppm with an average of 8.434 ppm. This value has exceeded the predetermined standard. The concentration of Cu in the leaves at location 1 (DTP-1) was exceedingly high compared to other samples because at location 1 it was contaminated with very high copper. This is in line with the results of the high magnetic susceptibility of the DTP-1 sample compared to other samples. High Cu content in the air can cause metal fever with symptoms such as flu-like symptoms, diarrhea, vomiting, irritation of the eyes, and dizziness [44].

The concentration of heavy metal Zn was detected in 20 leaf samples ranging from 1.087 ppm to 26.132 ppm with an average of 11.668 ppm. The concentration has exceeded the standard Zn in the air. High concentrations of Zn can be harmful to health including fatigue and weakness, anorexia, listlessness, headaches, chest pain, myalgia and arthralgia, gout, hypochromic microcytic anemia, and testicular atrophy [44]. Geochemical assessment of soils also can use to evaluate the source of Zn [47]. The dominant pollutant inputs are Fe, Cu, Zn and Pb respectively. Based on PLI calculations, most of the areas are affected by high to very high pollution levels.



**Table 2.** Heavy metals concentration on leaves determines using atomic absorption spectroscopy (AAS)

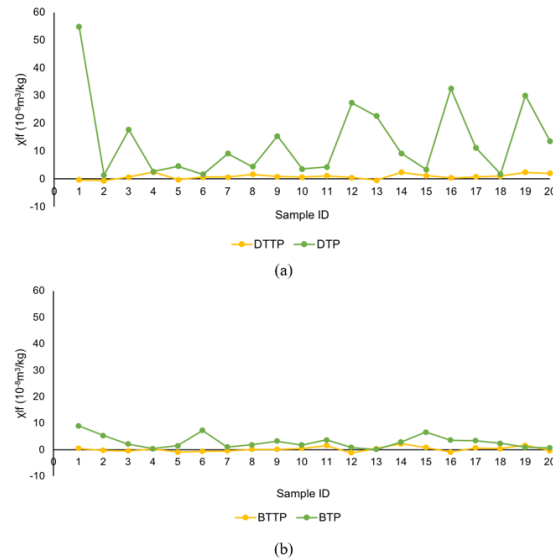
| Sample Code | $\chi_{lf}$ ( $10^{-8}$ m <sup>3</sup> /kg) | Heavy Metal Concentration (ppm) |         |         |         | Total Heavy Metal (ppm) | PLI    |
|-------------|---------------------------------------------|---------------------------------|---------|---------|---------|-------------------------|--------|
|             |                                             | Pb                              | Fe      | Cu      | Zn      |                         |        |
| REF         | -0.745                                      | 0.060                           | 1.016   | 0.057   | 1.543   |                         |        |
| DTP-1       | 54.764                                      | 0.138                           | 71.019  | 163.882 | 1.087   | 245.904                 | 23.913 |
| DTP-2       | 1.387                                       | 0.056                           | 1.731   | 0.323   | 7.008   | 9.118                   | 2.534  |
| DTP-3       | 17.741                                      | 0.104                           | 22.337  | 0.196   | 7.993   | 30.631                  | 5.118  |
| DTP-4       | 2.667                                       | 0.000                           | 0.550   | 0.093   | 2.273   | 2.916                   | 1.093  |
| DTP-5       | 4.536                                       | 0.060                           | 10.299  | 0.238   | 10.317  | 20.914                  | 4.109  |
| DTP-6       | 1.660                                       | 0.000                           | 1.212   | 0.166   | 3.992   | 5.369                   | 2.083  |
| DTP-7       | 9.158                                       | 0.075                           | 10.567  | 0.272   | 18.412  | 29.326                  | 5.225  |
| DTP-8       | 4.345                                       | 0.059                           | 9.895   | 0.288   | 11.831  | 22.073                  | 4.397  |
| DTP-9       | 15.408                                      | 0.000                           | 13.663  | 0.303   | 12.942  | 26.908                  | 8.451  |
| DTP-10      | 3.592                                       | 0.000                           | 8.535   | 0.270   | 8.194   | 16.998                  | 5.967  |
| DTP-11      | 4.259                                       | 0.055                           | 4.765   | 0.156   | 6.545   | 11.522                  | 2.668  |
| DTP-12      | 27.331                                      | 0.058                           | 18.185  | 0.309   | 17.123  | 35.676                  | 5.697  |
| DTP-13      | 22.656                                      | 0.058                           | 16.504  | 0.300   | 15.625  | 32.486                  | 5.388  |
| DTP-14      | 9.191                                       | 0.249                           | 13.492  | 0.332   | 19.997  | 34.070                  | 8.046  |
| DTP-15      | 3.349                                       | 0.123                           | 1.449   | 0.164   | 3.349   | 5.0855                  | 2.071  |
| DTP-16      | 32.440                                      | 0.000                           | 9.307   | 0.107   | 8.756   | 18.171                  | 4.620  |
| DTP-17      | 11.227                                      | 0.131                           | 13.813  | 0.298   | 13.348  | 27.591                  | 6.064  |
| DTP-18      | 1.6884                                      | 0.083                           | 2.632   | 0.307   | 19.218  | 22.239                  | 3.938  |
| DTP-19      | 30.003                                      | 0.065                           | 18.351  | 0.307   | 19.218  | 37.940                  | 6.021  |
| DTP-20      | 13.481                                      | 0.087                           | 18.131  | 0.367   | 26.132  | 44.718                  | 7.302  |
| Average     | 13.544                                      | 0.070                           | 13.322  | 8.434   | 11.668  |                         |        |
| Threshold   |                                             | **0.0005                        | *0.4375 | **0.001 | **0.005 |                         |        |
| R           |                                             | 0.1418                          | 0.8496  | 0.6984  | 0.2371  | 0.779                   |        |

Note: PLI, Pollution Load Index; REF, reference sample; DTP, polluted leaf samples. \*SE-01/MEN/1997 [48]; \*\*Occupational Safety and Health Administration (OSHA) [49].

#### 4 Discussion

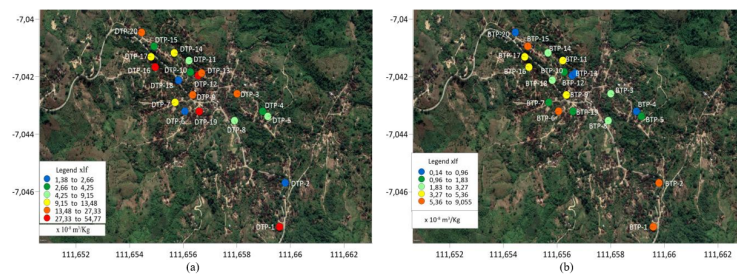
The average of magnetic susceptibility increases from non-polluted to polluted sample. Increasing of magnetic susceptibility of polluted samples in both leaf and bark, show that the leaf more sensitive than bark in record the pollution,  $\chi_{leaf} > \chi_{bark}$ . These results are supported by previous research conducted on Aztec Marigold, the shoots (leaves) of plants absorb more heavy metals Cd and Zn than the roots [50]. Leaf and bark magnetic susceptibility were also lower than other studies because controlled by the type of pollutant from hydrocarbon mining, which was different from general industrial pollutants. Besides that, *Muntingia calabura* has a smooth leaf and bark surface compared to other types of plants in previous studies, so that it is possible to reduce dust/pollutant trapping capacity. The increase in magnetic susceptibility between leaves and barks showed the same trend and illustrated in Figure 3. The increasing of magnetic susceptibility deduced due to increasing of heavy metals absorbed of iron (Fe and Cu) elements.

Figure 3 shows the increase in low frequency magnetic susceptibility ( $\chi_{lf}$ ) from non-polluted to polluted leaves and barks. The difference in average of magnetic susceptibility non polluted and polluted leaves and barks, it can calculate that the sensitivity of leaves is higher than that of barks with an increase in the average of magnetic susceptibility in leaves of 93.68% and bark of 82.83%. This results is in line with previous findings that leaves can be a better bioindicator of pollution than other plant organs [43]. The higher sensitivity of leaves compared to the barks in recording pollution can be understood, because leaves is related to the free air directly, while the barks are covered by leaves that are positioned outward and catch the pollutants first. Pollution that sticks to the barks is through the small space between leaves, so it is possible that the number of pollutants caught in the barks is not as much as in the leaves. In some point, if the increases look extreme, indicated that the contribution of pollutant is quite high so that it is possible to have a relation to mining production. The significant difference between the increase of magnetic susceptibility in leaf and barks also can be used as an indicator of leaf illness. Both assumptions require further verification. The 2D mapping of leaf and trunks magnetic susceptibility shown in Figure 4.



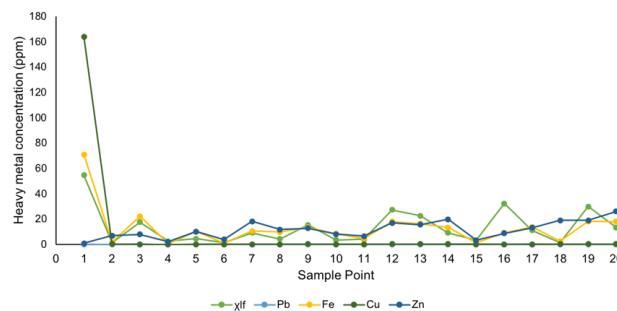
**Figure 3.** Trend of increasing magnetic susceptibility on (a) leaf and (b) bark of *Muntingia calabura* that spread on hydrocarbon mining area in Wonocolo Geopark

Note: DTTP, non-polluted leaf samples; DTP, polluted leaf samples; BTTP, non-polluted bark samples; BTP, polluted bark samples.



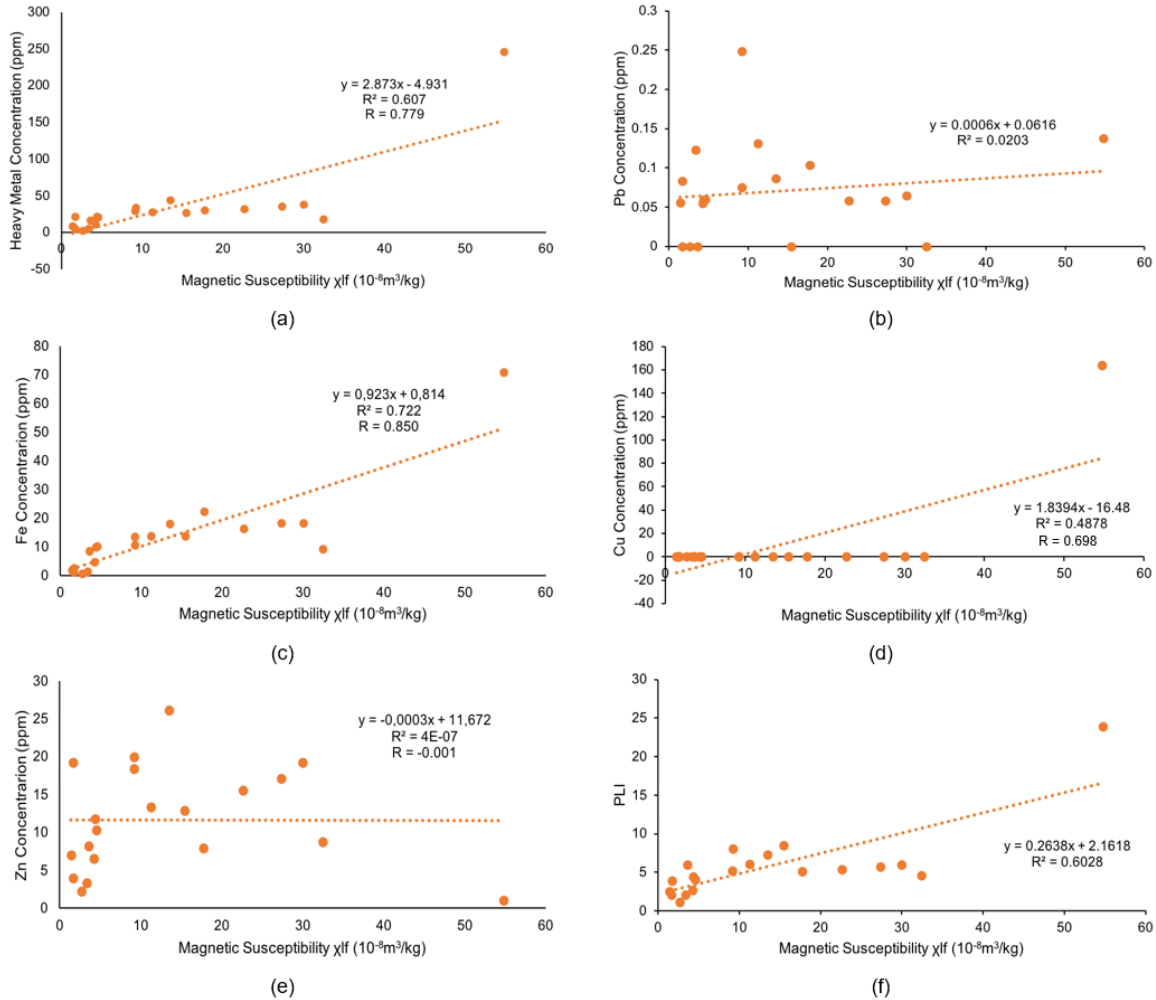
**Figure 4.** (a) Magnetic susceptibility distribution of polluted leaves and (b) magnetic susceptibility distribution of polluted barks

Figure 4 shows the distribution of magnetic susceptibility polluted leaves and barks in Wonocolo hydrocarbon mining site. Figure 4a for polluted leaves and Figure 4b for polluted barks, respectively. There is a difference in the distribution of the magnetic susceptibility. The magnetic susceptibility in almost all sampling points 1 to 20 of leaves is greater than that of bark. Earlier studies represent that increasing of susceptibility indicates the presence of heavy metal contributions in the air recorded by the leaves. Heavy metals found in the air in metropolitan cities include Zn, Fe, Pb, Ni, Cd, Cr [51]. The trend of magnetic susceptibility and heavy metal in every sampling point is shown in Figure 5. The high Cu values at site 1 could be due to the accumulation of Cu in leaves that have been exposed to pollution for a longer period of time, as well as the high level of mining activity at the site. It's also possible that the mining machinery at that location releases a significant amount of Cu. Pearson's correlation between the elemental content and PLI to the magnetic susceptibility is shown in Figure 6.



**Figure 5.** Heavy metal concentrations and magnetic susceptibility of leaves samples in every sampling point

The fluctuations in the magnetic susceptibility and heavy metals (Pb, Fe, Cu and Zn) have the same pattern and show significant correlation between magnetic susceptibility and heavy metals, mainly for total heavy metal. The Pearson's correlation of  $\chi$  vs Pb,  $\chi$  vs Fe,  $\chi$  vs Cu,  $\chi$  vs Zn and,  $\chi$  vs PLI shown in Figure 6. The concentration of heavy metals in leaves represents the concentration of heavy metals in the air. Table 2 shows the average concentration of heavy metal elements to be compared with the threshold of heavy metal in the air. All the heavy metals in leaf higher than that of the threshold. Iron element most the highest heavy element.



**Figure 6.** Relation between total heavy metal concentrations and low-frequency magnetic susceptibility ( $\chi_{lf}$ ): (a) Total heavy metals vs  $\chi_{lf}$  (Pearson's  $R = 0.0779$ ); (b) Pb vs  $\chi_{lf}$ ; (c) Fe vs  $\chi_{lf}$ ; (d) Cu vs  $\chi_{lf}$ ; (e) Zn vs  $\chi_{lf}$ ; and (f) Pollution Load Index (PLI) vs  $\chi_{lf}$

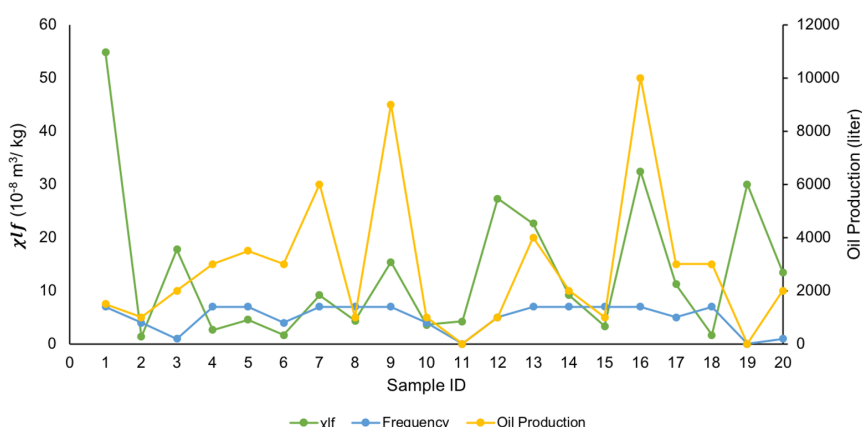
According to Occupational Safety and Health Administration (OSHA), the concentration of heavy metals in the air is above 8 working hours for Pb (0.0005 ppm), Cu (0.001 ppm), Zn (0.005 ppm) [44]. Meanwhile, according to the Minister of Environment, the threshold for Fe metal in the air is 0.4375 ppm. Based on these data, the anomalies of the four elements have exceeded the threshold for heavy metals in the air. The anomaly indicates the presence of anthropogenic contaminants which thought derived from machine and crude hydrocarbon refining activities.

The correlation between magnetic susceptibility and the total concentration of heavy metals shown in Figure 6. The value of magnetic susceptibility with the total concentration of heavy metals measured has a correlation coefficient (R) of 0.779. These results indicate that there is a meaningful relationship between the increase in magnetic susceptibility with an increase in the total concentration of heavy metals.

The relationship between magnetic susceptibility values and each heavy metal concentration shown in Figure 6. Heavy metal concentrations that have a strong relationship with magnetic susceptibility values are Fe and Cu metals with R-values of 0.850 and 0.698. While for heavy metals Pb and Zn have a very weak relation with magnetic susceptibility. The same case also in correlation with iron oxide and PM at atmosphere that have a negative correlation [52]. Fe is ferromagnetic, while the other three elements are diamagnetic. However, the formation of chalcophyte ( $\text{CuFeS}$ ) is possible, so the correlation between Fe and Cu and susceptibility can be positive. This

has also been confirmed in previous studies. Several previous studies have also shown that magnetic susceptibility correlates well with Fe and Cu and negatively with Pb and Zn. In some samples, Pb content found zero level, this was also mentioned by previous researchers where the leaves were not suitable as a biomonitor for Cl, K, P, S, As, Sc, Cs, Pb, Sn and Sr. As the recommendation previous scientist that a special plant serve a record for specific condition pollution such as highway, industrial area, so through this result of this study, we recommend that *Muntingia calabura* suitable serve biomonitoring in mining area [53].

Based on the prominent level of heavy metals cached by the leaves and the significance correlation between Fe and Cu with magnetic susceptibility, we can see an increasing trend of magnetic susceptibility with mining frequency and total production per week. The high frequency of mining and total production causes prominent levels of pollutants released by each production well. So, by understanding the trend of increasing certain pollutants which can be an indicator, the frequency of mining and production results described in Figure 7. Fe and Cu content can used as data on pollution indicators in mining areas due to mining frequency and the high production at the wells at that point. From the 20 observed points, it can be seen that the pattern of increasing and decreasing magnetic susceptibility in leaves is almost the same with increasing and decreasing mining frequency and total production. The increase in magnetic susceptibility is thought to be due to the input of magnetic minerals contained in the ash released by the diesel engine which is used as a driving force for hydrocarbon well pistons in exploration process. The higher the magnetic susceptibility, the higher the PLI and at a certain level of magnetic susceptibility, the PLI become saturated. If we refer to the input from the input pollutant type elements, it can be clearly seen that the most dominant input is Fe followed by Cu, Zn and Pb.



**Figure 7.** Distribution of magnetic susceptibility, mining frequency and total weekly production at each location point

## 5 Conclusions

In hydrocarbon mining site, the *Muntingia calabura*'s organ (leaf and bark) can record the pollution levels or serve as bioindicator of pollution or proxy of mining process and production. The record shown by increasing of susceptibility in leaf and barks for polluted samples with high level of heavy metals content (Pb, Fe, Cu and Zn). The dominant pollutant input is Fe followed by Cu, Zn and Pb. Magnetic susceptibility of polluted leaf, greater than that of bark,  $\chi_{\text{leaf}} > \chi_{\text{bark}}$ . The presence of heavy metals exceeds the threshold limit and based on PLI, the area have very high contamination. The significant correlation shows between the total heavy metals in the leaves and the magnetic susceptibility and also between magnetic susceptibility with Fe and Cu elements. The increase in magnetic susceptibility also followed by an increase in mining frequency and total mining production. Based on these results, the magnetic susceptibility of leaves and barks can used as a proxy indicators of air pollution levels and production levels of mining. Although the hydrocarbon is non-magnetic, the source of pollutants in the hydrocarbon mining area are contributed by iron oxide minerals from the combustion of diesel engines which are used to drive pistons in and out of wells and during the hydrocarbon refining process. Last but not less, this study suggest that the kind of trees used in this study, due to the special characteristic of fur leaves, *Muntingia calabura* is suitable as bioindicator of pollution and propose for greening solution tool of polluted area such as mining, industrial and city area.

## Author Contributions

Conceptualization, S.Z. and D.N.A.; methodology, S.Z., D.N.A., and A.L.; software, M.B.S.R., H.‘I.Z.; validation, C.A.H., A.T. and A.M.H.; formal analysis, D.N.A.; investigation, D.N.A. and A.L.; resources, S.Z.; data curation, S.Z. and D.N.A.; writing—original draft preparation, S.Z., H.‘I.Z., and M.F.R.H.; writing—review and editing, H.‘I.Z., C.A.H., A.M.H. and M.F.R.H.; visualization, M.B.S.R. and A.T.; supervision, S.Z.; project administration, D.N.A.; funding acquisition, S.Z. All authors have read and approved the final manuscript.



## Funding

Sampling and preparation sample at the first step of this research supported by funding from the LP2M Center Research Grant through the PNPB fund (Grant No.: 5.3.581/UN32.14.1/LT/2021) where S.Z. is the Chief Researcher. The addition of collecting sample and some measurement and analysis supported by Internal Funding of UM in 2022 (Grant No.: 19.5.994/UN32.20.1/LT/2022). More discussion and finishing this paper supported by Research Staff Exchange Funding from Universitas Negeri Malang 2024.

## Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

## Conflicts of Interest

The authors declare that they have no conflicts of interest.

## References

- [1] C. Zhang, B. Huang, J. D. A. Piper, and R. Luo, "Biomonitoring of atmospheric particulate matter using magnetic properties of *Salix matsudana* tree ring cores," *Sci. Total Environ.*, vol. 393, no. 1, pp. 177–190, 2008. <https://doi.org/10.1016/j.scitotenv.2007.12.032>
- [2] P. Gautam, U. Blaha, and E. Appel, "Magnetic susceptibility of dust-loaded leaves as a proxy of traffic-related heavy metal pollution in Kathmandu city, Nepal," *Atmos. Environ.*, vol. 39, no. 12, pp. 2201–2211, 2005. <https://doi.org/10.1016/j.atmosenv.2005.01.006>
- [3] J. Hofman, B. A. Maher, A. R. Muxworthy, K. Wuyts, A. Castanheiro, and R. Samson, "Biomagnetic monitoring of atmospheric pollution: A review of magnetic signatures from biological sensors," *Environ. Sci. Technol.*, vol. 51, no. 12, pp. 6648–6664, 2017. <https://doi.org/10.1021/acs.est.7b00832>
- [4] Q. Dai, M. Zhou, H. Li, X. Qian, M. Yang, and F. Li, "Biomagnetic monitoring combined with support vector machine: A new opportunity for predicting particle-bound-heavy metals," *Sci. Rep.*, vol. 10, no. 1, p. 8605, 2020. <https://doi.org/10.1038/s41598-020-65677-8>
- [5] D. Dawai, M. Macouin, S. Rousse, J. F. Léon, M. Gountié Dedzo, and L. Drigo, "Tracking airborne pollution with environmental magnetism in a medium-sized African city," *Atmosphere*, vol. 12, no. 10, p. 1281, 2021. <https://doi.org/10.3390/atmos12101281>
- [6] D. Muñoz, B. Aguilar, R. Fuentealba, and M. Préndez, "Environmental studies in two communes of Santiago de Chile by the analysis of magnetic properties of particulate matter deposited on leaves of roadside trees," *Atmos. Environ.*, vol. 152, pp. 617–627, 2017. <https://doi.org/10.1016/j.atmosenv.2016.12.047>
- [7] F. Kardel, K. Wuyts, K. De Wael, and R. Samson, "Biomonitoring of atmospheric particulate pollution via chemical composition and magnetic properties of roadside tree leaves," *Environ. Sci. Pollut. Res.*, vol. 25, no. 26, pp. 25 994–26 004, 2018. <https://doi.org/10.1007/s11356-018-2592-z>
- [8] H. Chen, D. S. Xia, B. Wang, H. Liu, and X. Ma, "Pollution monitoring using the leaf-deposited particulates and magnetism of the leaves of 23 plant species in a semi-arid city, Northwest China," *Environ. Sci. Pollut. Res.*, vol. 29, no. 23, pp. 34 898–34 911, 2022. <https://doi.org/10.1007/s11356-021-16686-1>
- [9] A. A. El-Khatib, N. A. Barakat, N. A. Youssef, and N. A. Samir, "Bioaccumulation of heavy metals air pollutants by urban trees," *Int. J. Phytoremediat.*, vol. 22, no. 2, pp. 210–222, 2020. <https://doi.org/10.1080/15226514.2019.1652883>
- [10] P. Sanderson, J. M. Delgado-Saborit, and R. M. Harrison, "A review of chemical and physical characterisation of atmospheric metallic nanoparticles," *Atmos. Environ.*, vol. 94, pp. 353–365, 2014. <https://doi.org/10.1016/j.atmosenv.2014.05.023>
- [11] M. D. Hays, S. H. Cho, R. Baldauf, J. J. Schauer, and M. Shafer, "Particle size distributions of metal and non-metal elements in an urban near-highway environment," *Atmos. Environ.*, vol. 45, no. 4, pp. 925–934, 2011. <https://doi.org/10.1016/j.atmosenv.2010.11.010>
- [12] N. V. Jordanova, D. V. Jordanova, L. Veneva, K. Yorova, and E. Petrovsky, "Magnetic response of soils and vegetation to heavy metal pollution: A case study," *Environ. Sci. Technol.*, vol. 37, no. 19, pp. 4417–4424, 2003. <https://doi.org/10.1021/es0200645>
- [13] A. Kapička, E. Petrovský, S. Ustjak, and K. Macháček, "Proxy mapping of fly-ash pollution of soils around a coal-burning power plant: A case study in the Czech Republic," *J. Geochem. Explor.*, vol. 66, no. 1–2, pp. 291–297, 1999. [https://doi.org/10.1016/S0375-6742\(99\)00008-4](https://doi.org/10.1016/S0375-6742(99)00008-4)
- [14] F. Heller, Z. Strzyszczyk, and T. Magiera, "Magnetic record of industrial pollution in forest soils of Upper Silesia, Poland," *J. Geophys. Res. Solid Earth*, vol. 103, no. B8, pp. 17 767–17 774, 1998. <https://doi.org/10.1029/98JB01667>

- [15] P. J. Flanders, "Collection, measurement, and analysis of airborne magnetic particulates from pollution in the environment (invited)," *J. Appl. Phys.*, vol. 75, no. 10, pp. 5931–5936, 1994. <https://doi.org/10.1063/1.355518>
- [16] World Health Organization, "Ambient (outdoor) air pollution," 2022. [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)
- [17] World Health Organization, "Who ambient air quality database (update 2023)," 2023. <https://www.who.int/publications/m/item/who-ambient-air-quality-database-%28update-2023%29>
- [18] L. Calderón-Garcidueñas, A. González-Maciel, P. S. Mukherjee, R. Reynoso-Robles, B. Pérez-Guillé, C. Gayosso-Chávez, R. Torres-Jardón, J. V. Cross, I. A. M. Ahmed, V. V. Karloukovski, and B. A. Maher, "Combustion- and friction-derived magnetic air pollution nanoparticles in human hearts," *Environ. Res.*, vol. 176, p. 108567, 2019. <https://doi.org/10.1016/j.envres.2019.108567>
- [19] T. Gonet and B. A. Maher, "Airborne, vehicle-derived Fe-bearing nanoparticles in the urban environment: A review," *Environ. Sci. Technol.*, vol. 53, no. 17, pp. 9970–9991, 2019. <https://doi.org/10.1021/acs.est.9b01505>
- [20] B. A. Maher, I. A. M. Ahmed, V. Karloukovski, D. A. MacLaren, P. G. Foulds, D. Allsop, D. M. A. Mann, R. Torres-Jardón, and L. Calderon-Garciduenas, "Magnetite pollution nanoparticles in the human brain," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 113, no. 39, pp. 10 797–10 801, 2016. <https://doi.org/10.1073/pnas.1605941113>
- [21] M. Sobolewski, K. Conrad, E. Marvin, M. Eckard, C. M. Goeke, A. K. Merrill, K. Welle, B. P. Jackson, R. Gelein, D. Chalupa, G. Oberdörster, and D. A. Cory-Slechta, "The potential involvement of inhaled iron (Fe) in the neurotoxic effects of ultrafine particulate matter air pollution exposure on brain development in mice," *Part. Fibre Toxicol.*, vol. 19, no. 1, p. 56, 2022. <https://doi.org/10.1186/s12989-022-00496-5>
- [22] D. Cory-Slechta, M. Sobolewski, and G. Oberdörster, "Air pollution-related brain metal dyshomeostasis as a potential risk factor for neurodevelopmental disorders and neurodegenerative diseases," *Atmosphere*, vol. 11, no. 10, p. 1098, 2020. <https://doi.org/10.3390/atmos11101098>
- [23] M. Riediker, D. Zink, W. Kreyling, G. Oberdörster, A. Elder, U. Graham, I. Lynch, A. Duschl, G. Ichihara, S. Ichihara, T. Kobayashi, N. Hisanaga, M. Umezawa, T. J. Cheng, R. Handy, M. Gulumian, S. Tinkle, and F. Cassee, "Particle toxicology and health - where are we?" *Part. Fibre Toxicol.*, vol. 16, no. 1, p. 19, 2019. <https://doi.org/10.1186/s12989-019-0302-8>
- [24] J. L. Allen, G. Oberdorster, K. Morris-Schaffer, C. Wong, C. Klocke, M. Sobolewski, K. Conrad, M. Mayer-Proschel, and D. A. Cory-Slechta, "Developmental neurotoxicity of inhaled ambient ultrafine particle air pollution: Parallels with neuropathological and behavioral features of autism and other neurodevelopmental disorders," *NeuroToxicology*, vol. 59, pp. 140–154, 2017. <https://doi.org/10.1016/j.neuro.2015.12.014>
- [25] M. Szönyi, L. Sagnotti, and A. M. Hirt, "On leaf magnetic homogeneity in particulate matter biomonitoring studies," *Geophys. Res. Lett.*, vol. 34, no. 6, p. L06306, 2007. <https://doi.org/10.1029/2006GL029076>
- [26] M. A. E. Chaparro, M. A. E. Chaparro, A. G. Castañeda Miranda, D. C. Marié, J. D. Gargiulo, J. M. Lavernia, M. Natal, and H. N. Böhnelt, "Fine air pollution particles trapped by street tree barks: In situ magnetic biomonitoring," *Environ. Pollut.*, vol. 266, p. 115229, 2020. <https://doi.org/10.1016/j.envpol.2020.115229>
- [27] T. C. L. Moreira, L. F. Amato-Lourenço, G. T. da Silva, C. D. Saldiva de André, P. A. de André, L. V. Barrozo, J. M. Singer, P. H. N. Saldiva, M. Saiki, and G. M. Locosselli, "The use of tree barks to monitor traffic related air pollution: A case study in São Paulo–Brazil," *Front. Environ. Sci.*, vol. 6, p. 72, 2018. <https://doi.org/10.3389/fenvs.2018.00072>
- [28] G. Kletetschka, "Magnetic anomalies on the tree trunks," *Stud. Geophys. Geod.*, vol. 47, no. 2, pp. 371–379, 2003. <https://doi.org/10.1023/A:1023779826177>
- [29] L. C. Vezzola, G. Muttoni, M. Merlini, N. Rotiroti, L. Pagliardini, A. M. Hirt, and M. Pelfini, "Investigating distribution patterns of airborne magnetic grains trapped in tree barks in Milan, Italy: Insights for pollution mitigation strategies," *Geophys. J. Int.*, vol. 210, no. 2, pp. 989–1000, 2017. <https://doi.org/10.1093/gji/ggx232>
- [30] P. Ballikaya, J. Marshall, and P. Cherubini, "Can tree-ring chemistry be used to monitor atmospheric nanoparticle contamination over time?" *Atmos. Environ.*, vol. 268, p. 118781, 2022. <https://doi.org/10.1016/j.atmosenv.2021.118781>
- [31] A. Winkler, C. Caricchi, M. Guidotti, M. Owczarek, P. Macrì, M. Nazzari, A. Amoroso, A. Di Giosa, and S. Listrani, "Combined magnetic, chemical and morphoscopic analyses on lichens from a complex anthropic context in Rome, Italy," *Sci. Total Environ.*, vol. 690, pp. 1355–1368, 2019. <https://doi.org/10.1016/j.scitotenv.2019.06.526>
- [32] N. Jordanova, D. Jordanova, E. Tcherkezova, B. Georgieva, and D. Ishlyanski, "Advanced mineral magnetic and geochemical investigations of road dusts for assessment of pollution in urban areas near the largest copper smelter in SE Europe," *Sci. Total Environ.*, vol. 792, p. 148402, 2021. <https://doi.org/10.1016/j.scitotenv.2021.148402>
- [33] S. Ayoubi, M. J. Samadi, H. Khademi, M. Shirvani, and Y. Gyasi-Agyei, "Using magnetic susceptibility for predicting hydrocarbon pollution levels in a petroleum refinery compound in Isfahan Province, Iran," *J. Appl. Geophys.*, vol. 172, p. 103906, 2020. <https://doi.org/10.1016/j.jappgeo.2019.103906>

- [34] K. L. Kapper, F. Bautista, A. Goguitchaishvili, M. F. Bógalo, R. Cejudo-Ruíz, and M. C. Solano, "The use and misuse of magnetic methods to monitor environmental pollution in urban areas," *Bol. Soc. Geol. Mex.*, vol. 72, no. 1, pp. 1–44, 2020. <https://doi.org/10.18268/BSGM2020v72n1a111219>
- [35] B. Wang, D. Xia, Y. Yu, J. Jia, and S. Xu, "Detection and differentiation of pollution in urban surface soils using magnetic properties in arid and semi-arid regions of northwestern China," *Environ. Pollut.*, vol. 184, pp. 335–346, 2014. <https://doi.org/10.1016/j.envpol.2013.08.024>
- [36] T. Magiera, B. Górka-Kostrubiec, T. Szumiata, and M. Wawer, "Technogenic magnetic particles from steel metallurgy and iron mining in topsoil: Indicative characteristic by magnetic parameters and Mössbauer spectra," *Sci. Total Environ.*, vol. 775, p. 145605, 2021. <https://doi.org/10.1016/j.scitotenv.2021.145605>
- [37] B. H. Iswanto, I. Pratiwi, and S. Zulaikah, "Identification of environments based on magnetic susceptibility and geochemical data using multivariate statistical analysis," in *International Conference on Electromagnetism, Rock Magnetism and Magnetic Material (ICE-R3M)*, Malang, Indonesia, 2020, p. 040005. <https://doi.org/10.1063/5.0015862>
- [38] I. Szczepaniak-Wnuk, B. Górka-Kostrubiec, S. Dytłow, P. Szwarczewski, P. Kwapuliński, and J. Karasiński, "Assessment of heavy metal pollution in Vistula River (Poland) sediments by using magnetic methods," *Environ. Sci. Pollut. Res.*, vol. 27, no. 19, pp. 24 129–24 144, 2020. <https://doi.org/10.1007/s11356-020-08608-4>
- [39] E. Kelepertzis, A. Argyraki, F. Botsou, E. Aidona, A. Szabó, and C. Szabó, "Tracking the occurrence of anthropogenic magnetic particles and potentially toxic elements (PTEs) in house dust using magnetic and geochemical analyses," *Environ. Pollut.*, vol. 245, pp. 909–920, 2019. <https://doi.org/10.1016/j.envpol.2018.11.072>
- [40] M. L. Rijal, K. Porsch, E. Appel, and A. Kappler, "Magnetic signature of hydrocarbon-contaminated soils and sediments at the former oil field Hänigsen, Germany," *Stud. Geophys. Geod.*, vol. 56, no. 3, pp. 889–908, 2012. <https://doi.org/10.1007/s11200-010-0040-8>
- [41] F. Karimian, S. Ayoubi, B. Khalili, and S. A. Mireei, "Magnetic susceptibility as a proxy for detection of total petroleum hydrocarbons in contaminated wetlands," *Environ. Monit. Assess.*, vol. 195, no. 1, p. 244, 2023. <https://doi.org/10.1007/s10661-022-10826-x>
- [42] J. T. Lin, Y. Y. Chang, Y. C. Chen, L. C. Kuo, and D. J. Yang, "Protective effect and mechanism of Muntingia calabura Linn. fruit ethanolic extract against vascular endothelial growth factor production in nickel-stimulated hepatocellular carcinoma cells," *J. Funct. Foods*, vol. 47, pp. 343–349, 2018. <https://doi.org/10.1016/j.jff.2018.06.001>
- [43] E. Govea-Alcaide, S. H. Masunaga, A. De Souza, L. Fajardo-Rosabal, F. B. Effenberger, L. M. Rossi, and R. F. Jardim, "Tracking iron oxide nanoparticles in plant organs using magnetic measurements," *J. Nanopart. Res.*, vol. 18, no. 10, p. 305, 2016. <https://doi.org/10.1007/s11051-016-3610-z>
- [44] J. Briffa, E. Sinagra, and R. Blundell, "Heavy metal pollution in the environment and their toxicological effects on humans," *Heliyon*, vol. 6, no. 9, p. e04691, 2020. <https://doi.org/10.1016/j.heliyon.2020.e04691>
- [45] S. V. Torti and F. M. Torti, "Iron and cancer: More ore to be mined," *Nat. Rev. Cancer*, vol. 13, no. 5, pp. 342–355, 2013. <https://doi.org/10.1038/nrc3495>
- [46] P. Chen, M. Totten, Z. Zhang, H. Bucinca, K. Erikson, A. Santamaría, A. B. Bowman, and M. Aschner, "Iron and manganese-related CNS toxicity: Mechanisms, diagnosis and treatment," *Expert Rev. Neurother.*, vol. 19, no. 3, pp. 243–260, 2019. <https://doi.org/10.1080/14737175.2019.1581608>
- [47] G. N. Salomão, D. d. L. Farias, P. K. Sahoo, R. Dall'Agnol, and D. Sarkar, "Integrated geochemical assessment of soils and stream sediments to evaluate source-sink relationships and background variations in the Parauapebas River Basin, eastern Amazon," *Soil Syst.*, vol. 5, no. 1, p. 21, 2021. <https://doi.org/10.3390/soilsystems5010021>
- [48] Menteri Tenaga Kerja Republik Indonesia, "Surat Edaran Menteri Tenaga Kerja No. SE-01/MEN/1997 Tentang Nilai Ambang Batas Faktor Kimia Di Udara Lingkungan Kerja," 1997. <https://www.scribd.com/document/70855301/Surat-Edaran-Menaker-No-Se-01-Men-1997>
- [49] R. Z. Witter, L. Tenney, S. Clark, and L. S. Newman, "Occupational exposures in the oil and gas extraction industry: State of the science and research recommendations," *Am. J. Ind. Med.*, vol. 57, no. 7, pp. 847–856, 2014. <https://doi.org/10.1002/ajim.22316>
- [50] M. T. Madanan, I. K. Shah, G. K. Varghese, and R. K. Kaushal, "Application of Aztec Marigold (*Tagetes erecta* L.) for phytoremediation of heavy metal polluted lateritic soil," *Environ. Chem. Ecotoxicol.*, vol. 3, pp. 17–22, 2021. <https://doi.org/10.1016/j.enceco.2020.10.007>
- [51] P. R. Chaudhari, R. Gupta, D. G. Gajghate, and S. R. Wate, "Heavy metal pollution of ambient air in Nagpur City," *Environ. Monit. Assess.*, vol. 184, no. 4, pp. 2487–2496, 2012. <https://doi.org/10.1007/s10661-011-2133-4>
- [52] E. Petrovský, A. Kapička, H. Grison, B. Kotlík, and H. Miturová, "Negative correlation between concentration of iron oxides and particulate matter in atmospheric dust: Case study at industrial site during smoggy period," *Environ. Sci. Eur.*, vol. 32, no. 1, p. 134, 2020. <https://doi.org/10.1186/s12302-020-00420-8>

- [53] A. Schmidt, R. Yarnold, M. Hill, and M. Ashmore, “Magnetic susceptibility as proxy for heavy metal pollution: A site study,” *J. Geochem. Explor.*, vol. 85, no. 3, pp. 109–117, 2005. <https://doi.org/10.1016/j.gexplo.2004.12.001>

## Nomenclature

BTP    Bark samples  
DTP    Leaf samples  
BTTP   Non-polluted bark samples  
DTTP   Non-polluted leaf samples  
OSHA   Occupational Safety and Health Administration  
PLI    Pollution Load Index  
ppm    Parts per Million

## Greek symbols

$\chi$        Magnetic susceptibility,  $10^{-8} \text{ m}^3/\text{kg}$   
 $\chi_{lf}$     Magnetic susceptibility (low-frequency),  $10^{-8} \text{ m}^3/\text{kg}$